

RCM-Constrained Impedance MPC for Force-Safe Robotic Laparoscopy

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Abstract—Robot-assisted laparoscopy must steer an instrument that pivots about the trocar incision—the remote center of motion (RCM)—while keeping tool–tissue forces safe under nonlinear tissue contact and respiration. We extend a two-layer Impedance MPC in which a feedforward layer cancels manipulator nonlinearities, exposing a configuration-independent double-integrator residual, and a small convex QP with an augmented Kalman disturbance state delivers offset-free tracking. First, the RCM is formulated as a hard constraint at every horizon step (and analyzed; validation isolates the joint-space rows as the sole remaining component). Second, the random-walk disturbance model is augmented with an adaptive Hunt–Crossley tissue estimator and a respiration oscillator, so structured disturbances are predicted rather than merely rejected. Third, since the post-feedforward plant is linear, the feedback is synthesized directly: the H_2 gain is the classical impedance the unconstrained MPC realizes, while H_∞ bounds the disturbance-to-force gain by γ , making peak-force safety a closed-loop invariant. The constant- A_d structure is preserved, keeping the QP solvable in under 1 ms. A MuJoCo implementation on a 7-DoF manipulator with a compliant trocar verifies the design: steady-state tip error falls from 3.94 mm to 0.03 mm under sustained tissue contact, the force-safety governor caps a commanded tissue overrun near the 3 N budget versus 5.4 N push-through, and force safety survives sensor noise, bias, and stick-slip trocar friction.

Index Terms—Surgical robotics, impedance control, model predictive control, remote center of motion, Hunt–Crossley contact model, H_∞ control, disturbance rejection.

I. INTRODUCTION

ROBOT-ASSISTED minimally invasive surgery is a clinical standard, yet its core control problem remains open. A tool enters the abdomen through a trocar, which imposes a remote-center-of-motion (RCM) constraint: the tool axis must pass through a fixed incision point throughout the procedure, or lateral force tears the abdominal wall. Inside, the tool contacts soft tissue whose properties are nonlinear, viscoelastic, and patient-specific, while respiration (0.1–0.5 Hz), heartbeat, and hand tremor (8–12 Hz) inject persistent disturbances. A clinically usable controller must therefore (i) enforce the RCM, (ii) keep tool–tissue force compliant and bounded, (iii) reject physiological disturbances with a provable bound, and (iv) be robust to tissue-parameter uncertainty.

This paper builds on the two-layer Impedance MPC for physical human–robot interaction (pHRI) of Cao and Tang [1]. There, Layer 1 cancels gravity, Coriolis, and task-space inertia, reducing the plant to a configuration-independent double integrator with constant A_d ; Layer 2 solves a small

convex QP against it; an augmented Kalman filter gives a formal zero-steady-state-error guarantee; and a dual barrier enforces joint limits—attaining sub-0.05 mm steady-state error on a 7-DoF manipulator under a 15 N push, an 800-fold reduction over classical impedance. The constant- A_d structure, precomputed free response, Kalman augmentation, and impedance-equivalence result of [1] are inherited without re-derivation. The laparoscopic setting, however, forces three genuine extensions—a different *constraint* (the RCM is a task-space holonomic constraint, not a one-sided joint bound), a different *disturbance* (structured tissue contact, not a slow push), and a different *safety bar* (a pointwise peak-force bound, not only offset-free tracking)—each a contribution of the present paper.

A. Related Work

RCM enforcement. Robot-assisted laparoscopy has historically enforced the RCM *kinematically*, through a constrained-Jacobian formulation projecting the commanded tool-tip motion onto the subspace consistent with a stationary trocar point [11]; more recent schemes move enforcement to the *torque* level, treating the RCM as a holonomic constraint with a Lagrange multiplier and splitting the joint torque into constraint-maintaining and free-motion components via projection-based inverse dynamics [12]. Both families enforce the pivot only at the current instant—the predicted future trajectory is left unconstrained. Imposing the RCM *predictively*, as a hard constraint at every horizon step, is one of our contributions.

Predictive variable impedance. The closest line is model-predictive variable impedance control (MPVIC). Roveda *et al.* [2] combine a low-level Cartesian impedance loop, an MPC outer loop selecting time-varying stiffness and damping from an adaptively estimated Hunt–Crossley environment model, and a virtual energy tank restricting the MPC to passivity-preserving moves; Anand *et al.* [3] replace the analytic model with learned forward dynamics and cross-entropy stiffness sampling. We share MPVIC’s adaptive Hunt–Crossley estimator but differ in three structural ways inherited from [1]: (i) Layer-1 cancellation yields a configuration-independent A_d , eliminating the dominant cost of nonlinear-plant MPC and enabling 100 Hz operation with a small QP, versus the 10–30 Hz typical when the nonlinear plant is retained; (ii) safety is offset-free tracking plus a *peak-force* bound rather than energy-tank passivity—a force threshold, not net energy, is what tears tissue; and (iii) the impedance is not co-optimized with the trajectory—the unconstrained MPC is *provably* a

classical impedance whose realized gains are fixed by the cost weights, so the design space is the weights, not the gains.

H_∞ impedance and offset-free MPC. Larby and Forni [4] give an offline, passivity-preserving H_∞ synthesis of fixed impedance parameters via sparsity-constrained LMIs; we instead synthesize the feedback for the *post-feedforward linear plant* and embed it in the online MPC, so the realized gains co-vary with state, tissue estimate, and constraint activity. The offset-free principle—an integrating disturbance state removing steady-state error under persistent disturbance [5], [15], with the saturated-linear case in [6]—underlies the inherited Kalman augmentation, extended here to a *structured* (Hunt–Crossley) disturbance by an adaptive parametric estimator running in parallel. A constant- A_d predictive architecture, a hard RCM constraint, adaptive Hunt–Crossley estimation, and an H_∞ peak-force design have not previously been combined.

B. Contributions

- 1) **RCM as a hard predictive constraint** (Section III): we cast the trocar pivot as a rheonomic holonomic constraint imposed at every horizon step, replacing the joint-limit barrier of [1] while preserving the constant- A_d structure. We contribute the design and analysis; the gain-insensitivity result (T3, Section VI-C) isolates the decision-coupled joint-space rows as necessary for the sub-0.5 mm band.
- 2) **Adaptive Hunt–Crossley tissue estimator** (Section IV): a dead-zone recursive-least-squares law runs in parallel with the Kalman filter to keep the predicted force calibrated as tissue stiffness varies, while a respiration oscillator makes the dominant periodic disturbance *predicted* rather than merely rejected.
- 3) H_∞/H_2 **design for force safety** (Section V): since Layer-1 cancellation leaves a linear plant, the feedback is synthesized directly—the H_2 gain coincides with the impedance realized by the unconstrained MPC, and H_∞ bounds the disturbance-to-force gain by γ . With the MPC as its constrained realization, γ sets the budget $F_{\max} = \gamma \bar{w}$, making peak-force safety a design-level invariant (Proposition 1, Theorem 1).
- 4) **Measured verification** (Section VI): a MuJoCo implementation verifies the offset-free tracking, the adaptive estimator, and the force-safety governor under sensor noise, bias, stick-slip friction, respiration mismatch, and latency—isolating the joint-space hard-RCM rows as the sole remaining component.

II. LAPAROSCOPIC SYSTEM DYNAMICS

A. Manipulator Dynamics

The manipulator equation of motion of [1] is retained:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + J^\top(q)F_{\text{ext}}, \quad (1)$$

where $F_{\text{ext}} \in \mathbb{R}^6$ is now the *tool–tissue* wrench rather than the human-applied wrench. The 7-DoF manipulator carries a 30 cm rigid laparoscopic instrument; the dynamic structure and the operational-space inertia $\Lambda(q) = (J_v M^{-1} J_v^\top)^{-1}$ [8] are unchanged.

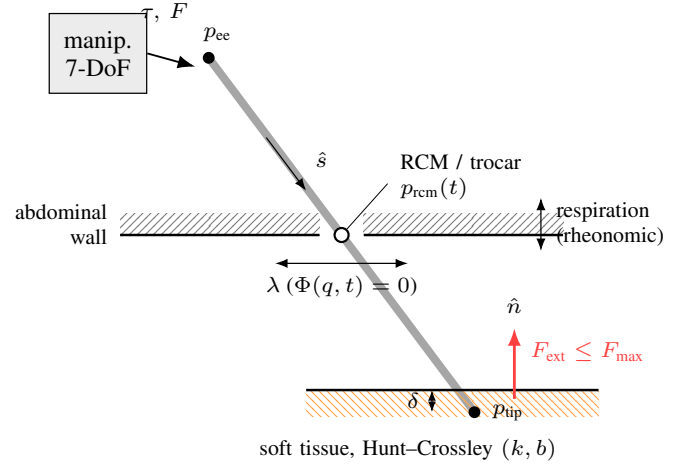


Fig. 1. RCM-constrained laparoscopic plant, mapping the variables of this section onto the physical parts: a 7-DoF manipulator applies joint torque τ and tip force F (1) through a rigid instrument with flange p_{ee} , tip p_{tip} , and shaft direction $\hat{s}$. The instrument pivots about the trocar incision—the remote center of motion $p_{rcm}(t)$, a rheonomic point that drifts with respiration—where the lateral deviation $\Phi(q, t)$ (2) must be held at zero against the lateral trocar force λ . The tip presses on soft tissue modeled by Hunt–Crossley contact (stiffness k , damping b , penetration δ) (4), producing the contact force F_{ext} held under the hard peak-force safety bound $F_{\max}$ along the contact normal $\hat{n}$. Respiration, heartbeat, and tremor enter as the structured disturbance of Section II-D.

B. RCM as a Rheonomic Holonomic Constraint

Let $p_{rcm}(t) \in \mathbb{R}^3$ denote the trocar point, allowed to vary slowly as the patient breathes. With end-effector flange $p_{ee}(q)$, tool tip $p_{tip}(q)$, and shaft direction $\hat{s}(q) = (p_{tip} - p_{ee}) / \|p_{tip} - p_{ee}\|$, the perpendicular deviation of the trocar from the tool axis is

$$\Phi(q, t) = (p_{rcm} - p_{ee}) - [(p_{rcm} - p_{ee})^\top \hat{s}] \hat{s}, \quad (2)$$

and the RCM constraint is $\Phi(q, t) = 0$. Because $p_{rcm}(t)$ depends explicitly on time, this is a rheonomic holonomic constraint of dimension two. Differentiating, $\dot{\Phi} = J_c(q, t)\dot{q} + \partial\Phi/\partial t$ with $J_c = \partial\Phi/\partial q$, and the constrained dynamics become

$$M\ddot{q} + C\dot{q} + G = \tau + J_c^\top \lambda + J^\top F_{\text{ext}}, \quad (3)$$

with $\lambda \in \mathbb{R}^2$ the lateral trocar force. The RCM acts on q , not directly on the residual state of the predictive architecture; it is imposed via the kinematic chain $q_k \rightarrow (p_{tip, k}, p_{ee, k}) \rightarrow \Phi_k$ at every horizon step without affecting the constant- A_d structure.

C. Tissue Interaction: Hunt–Crossley Contact

Where [1] modelled the human push as an unknown bounded force absorbed by the integrating disturbance state, the laparoscopic interaction force has known structure. We adopt the Hunt–Crossley model [9], [10]:

$$F_{\text{ext}}(\delta, \dot{\delta}) = k_e \delta^n (1 + b \dot{\delta}), \quad \delta \geq 0, \quad (4)$$

with penetration depth δ , stiffness k_e , exponent $n \approx 1.5$ (assumed known), and damping b . Linear viscoelastic models (Maxwell, Kelvin–Voigt, Zener) fit soft biological tissue poorly—they predict discontinuous contact force and constant

damping; (4) gives continuous contact force and penetration-scaled damping, both supported empirically. The unknown parameters $\theta = [k_e, b]^\top$ are adapted online (Section IV).

The scalar δ is defined along the contact normal, not a fixed world axis: $\delta = \max(0, (p_{\text{tip}} - p_c)^\top \hat{n})$ and $\dot{\delta} = \dot{p}_{\text{tip}}^\top \hat{n}$, with contact point p_c and surface normal $\hat{n}$ from the contact solver. On a curved organ $\hat{n}$ rotates as the tool slides, so the Hunt–Crossley force is evaluated in the instantaneous contact frame with force direction $-\hat{n}$. Tissue heterogeneity is handled by the online estimator itself: θ is spatially varying, and the dead-zone RLS re-converges when the tool crosses a tissue boundary.

D. Disturbance Model

The lumped disturbance acting on the Cartesian residual dynamics decomposes as

$$d(t) = d_{\text{tissue}}(\delta, \dot{\delta}; \theta) + d_{\text{resp}}(t) + d_{\text{trem}}(t) + d_{\text{tro}}(t) + \epsilon_m(t), \quad (5)$$

where d_{tissue} is the predicted Hunt–Crossley term, handled by the tissue estimator; $d_{\text{resp}}, d_{\text{trem}}, d_{\text{tro}}$ are physiological and mechanical disturbances handled by the Kalman state—with the quasi-periodic respiratory component given a dedicated oscillator internal model (Section III-E); and ϵ_m is residual model error. The residual $d - d_{\text{tissue}}$ is what the Kalman filter sees, bounded by $\|d - d_{\text{tissue}}\|_2 \leq \bar{w}$ with $\bar{w}$ smaller than it would be without the parametric prediction.

This split is what makes the force-safety constraint (10c) well-posed. Although the contact force enters the error dynamics as a disturbance ($d_{\text{tissue}} \approx -F_{\text{ext}}$, force form), it is not an *exogenous* one: $F_{\text{ext}} = k_e \delta^n (1 + b\dot{\delta})$ is a state-dependent, controllable contact law—the penetration is a function of the commanded tool-tip position—so it is *predicted* from the planned trajectory and *shaped* by how the tool pushes; it can therefore be *constrained*. The genuinely exogenous remainder $w = d - d_{\text{tissue}}$ (friction, respiration residual, model error) is neither predictable nor controllable and is instead *rejected*—offset-free by the Kalman filter, attenuated by the H_∞ gain of Section V. A hard bound on F_{ext} would be ill-posed if it were an arbitrary exogenous disturbance; it is enforceable precisely because its dominant part is the structured contact law, which is why the adaptive estimator is a necessary ingredient rather than a refinement.

III. TWO-LAYER IMPEDANCE MPC WITH RCM CONSTRAINT

The architecture of [1] is inherited unchanged in its essential structure; this section states what is the same and what is new.

A. Layer 1: Feedforward Nonlinear Inversion

The Layer-1 torque decomposition of [1] is preserved:

$$\tau = \tau_{\text{ff}} + J_v^\top F_{\text{mpc}} + J_\omega^\top F_{\text{orient}} + \bar{N}^\top \tau_{\text{null}}, \quad (6)$$

with $\tau_{\text{ff}} = C\dot{q} + G + J_v^\top \Lambda \ddot{p}_d$. After cancellation the residual error dynamics are

$$\ddot{e} = -\Lambda^{-1}(q)F_{\text{mpc}} + d_{\text{res}}(t), \quad (7)$$

with d_{res} the residual after both Layer-1 cancellation and the parametric tissue prediction. Exact zero-order-hold discretization (the continuous state matrix is nilpotent) gives

$$x_{e,k+1} = \underbrace{\begin{bmatrix} I_3 & \Delta t I_3 \\ 0 & I_3 \end{bmatrix}}_{A_d \text{ (constant)}} x_{e,k} + \underbrace{\begin{bmatrix} 0 \\ -\Lambda^{-1}(q_k)\Delta t \end{bmatrix}}_{B_d(\rho_k)} F_{\text{mpc},k}, \quad (8)$$

with $x_e = [e^\top, \dot{e}^\top]^\top$. The plant is an LPV system with a *constant* state matrix A_d and a parameter-varying input matrix $B_d(\rho_k)$, the only configuration dependence entering through $\Lambda^{-1}(q_k)$; “constant- A_d ” is used throughout as shorthand for this structure. Its value is computational: the free-response matrix $\Phi_N = [A_d; A_d^2; \dots; A_d^N]$ and the stage-cost Hessian blocks are formed once offline, while only $B_d(\rho_k)$ —hence the input-to-state map Γ —is rebuilt online each step.

B. Layer 2: QP with RCM and Force-Safety Constraints

The QP of [1] is extended with the RCM constraint and the force-safety bound. A *short* horizon $N \in [5, 20]$ is used ($N = 10$ in the experiments): a longer horizon sharpens constraint look-ahead, but by the disturbance-prediction degradation of Section III-F it also worsens the predicted force and RCM quantities and inflates the force-budget back-off. With decision variables $U = [F_{\text{mpc},0}; \dots; F_{\text{mpc},N-1}] \in \mathbb{R}^{3N}$ (translational-only at the tool tip; orientation is handled by Layer 1),

$$\min_U \frac{1}{2} U^\top H U + h^\top U \quad (9)$$

subject to, for $k = 0, \dots, N-1$ (terminal index N),

$$\|\Lambda(q_k) \ddot{p}_{d,k} + F_{\text{mpc},k}\|_\infty \leq F_{\text{max},u}, \quad (10a)$$

$$\Phi(q_k, t_k) = 0, \quad k = 0, \dots, N, \quad (10b)$$

$$\|\hat{F}_{\text{ext},k}\|_\infty \leq F_{\text{max}}, \quad \hat{F}_{\text{ext},k} = \hat{k}_e \delta_k^n (1 + \hat{b} \dot{\delta}_k) + \hat{n}^\top \hat{d}_k, \quad (10c)$$

$$\xi_N \in \mathcal{X}_f, \quad (10d)$$

where $H = \Gamma^\top \bar{Q} \Gamma + \bar{R}$ and $h = \Gamma^\top \bar{Q} x_{\text{free}}$ exactly as in [1], with $x_{\text{free}} = \Phi_N x_e + D_{\text{bar}} \hat{d}$. The modifications are:

1) *Effort bound* (10a): bounds the *total* applied tip force—the feedforward inertial term plus the MPC correction—since that sum is what the manipulator exerts at the tip. $F_{\text{max},u}$ is a Cartesian tip-force effort ceiling chosen so the induced joint torques $\tau = J_v^\top F$ stay within the actuator envelope; because $\Lambda(q_k) \ddot{p}_{d,k}$ is known at solve time, (10a) is an affine shift of a box and preserves the QP structure. Operation is restricted to the singularity-free workspace, with $F_{\text{max},u}$ scaled online by the smallest singular value of J_v , and Layer 1 applies a torque-consistent *directional* saturation scaling as a hardware backstop—the entire task-space command is scaled down rather than clipped joint-by-joint, so $|\tau_i| \leq \tau_{\text{max},i}$ holds rigorously while the compliance direction is preserved. Inside the designated workspace this scaling never binds tightly enough to deprive the RCM rows of joint-torque authority.

2) *Hard RCM* (10b): is imposed over the whole horizon. A single linearization about q_0 is accurate only near the current pose; over a longer horizon the predicted configuration drifts and the constraint manifold’s curvature makes a one-point linearization mis-estimate the deviation. The constraint is therefore linearized *along the predicted trajectory* (LPV):

at horizon step i the rows use $J_c(\hat{q}_{k+i})$ evaluated at the configuration predicted by the previous QP solution (a one-sweep successive convexification). The residual linearization error is absorbed by tightening the band over the horizon, $\varepsilon_i = \varepsilon - \kappa i \Delta t$ with κ a curvature-dependent back-off made explicit by the following bound. Because A_d is unchanged, only these constraint rows are rebuilt online.

Lemma 1 (RCM linearization error). *Let $\Phi(\cdot, t_k)$ be twice continuously differentiable on the operating region, with curvature bound $L_\Phi := \sup_q \|\partial^2 \Phi / \partial q^2\|_2$, and let the one-sweep linearization Φ_{lin} be taken about the predicted configuration $\hat{q}_{k+i}$ of the previous QP solution. If the predicted trajectory stays within $\|q_{k+i} - \hat{q}_{k+i}\|_2 \leq v_q i \Delta t$ of that sweep, with v_q a bound on the inter-sweep joint-velocity change, then*

$$\|\Phi(q_{k+i}, t_k) - \Phi_{\text{lin}}(q_{k+i}, t_k)\|_2 \leq \frac{1}{2} L_\Phi (v_q i \Delta t)^2. \quad (11)$$

Proof: Taylor's theorem with second-order Lagrange remainder, applied componentwise to Φ about $\hat{q}_{k+i}$: the constant and first-order terms are Φ_{lin} and the remainder of each component is bounded by $\frac{1}{2} L_\Phi \|q_{k+i} - \hat{q}_{k+i}\|_2^2$; substituting the trajectory bound gives (11). ■

Equation (11) fixes the back-off rather than leaving it free: choosing $\kappa = \frac{1}{2} L_\Phi v_q^2 N \Delta t$ makes the linear tightening $\kappa i \Delta t$ dominate the quadratic error (11) for every $i \leq N$, so κ is a computable curvature constant, not a tuned quantity.

3) *Force safety* (10c): is the hard peak-force bound. $\hat{F}_{\text{ext},k}$ is the predicted contact force along the contact normal and combines the structured Hunt–Crossley term—evaluated from the planned trajectory and tissue estimate $\hat{\theta}$ —plus the contact-normal component $\hat{n}^\top \hat{d}_k$ of the Kalman disturbance estimate, so the constraint limits the controller's best current estimate of the total normal force, consistent with the H_∞ budget $F_{\text{max}} = \gamma \bar{w}$ of Section V in which $\bar{w}$ bounds only the prediction error of that estimate. The respiration component enters through the penetration δ_k (the predicted surface follows the oscillator forecast of Section III-E), and only the normal projection of $\hat{d}_k$ is added, so shaft-axial trocar friction tightens the bound conservatively. Over the horizon $\hat{F}_{\text{ext},k}$ is linearized along the predicted trajectory, exactly as the RCM rows.

4) *Terminal constraint* (10d): acts on the predicted augmented state $\xi = [x_e^\top, \hat{d}^\top, x_r^\top]^\top$ (Section III-E). $\mathcal{X}_f$ is a positively invariant set under the terminal feedback (the H_∞ /LQR gain of Section V) inside which (10b)–(10c) remain satisfiable; together with the terminal cost it yields recursive feasibility and closed-loop stability in the standard way [13]. It replaces the joint-limit dual barrier of [1].

Prediction chain. The decision U lives in Cartesian residual coordinates, so the configuration needed by (10b)–(10c) is recovered explicitly: the predicted tip is $p_{\text{tip},k+i} = p_{d,k+i} - e_{k+i}$ with e_{k+i} the position block of $\Phi_N x_{e,k} + \Gamma U_{k+i}$, and the predicted configuration is

$$\hat{q}_{k+i} = q_k + \bar{J}_v^+(q_k) (p_{\text{tip},k+i} - p_{\text{tip},k}) + \bar{N}(q_k) \Delta q_{\text{null},i},$$

with $\bar{J}_v^+$ the dynamically consistent pseudoinverse and $\Delta q_{\text{null},i}$ the redundant increment from the secondary RCM/posture law. Two regimes use this chain differently. In the *joint-space realization* required for a hard RCM, the decision is

lifted to the joint motion itself, so (10b) is a genuine linear row in the decision variables. In the *tip-space implementation* of Section VI, the decision is the Cartesian force and the redundant term is set by a null-space regulator outside the QP—which is exactly why RCM there is only soft, as the gain-insensitivity result (T3, Section VI-C) confirms.

Remark (why (10c) is an output constraint, and why it is needed). The contact force does not appear in the prediction model (8); there it acts only as the disturbance estimate $\hat{d}$ entering the free response. In (10c) it appears as a *constrained output*—a function of the predicted state and the tissue estimate via the kinematic chain $x_{e,k} \rightarrow p_{\text{tip},k} \rightarrow \delta_k$. Constraining an output absent from the dynamics is standard in MPC. It is *necessary*, not redundant with disturbance rejection, because the two treat F_{ext} with opposite intent: the offset-free layer *rejects* F_{ext} to drive the tip to p_d —if p_d lies inside tissue this means pushing through with whatever force the reference demands. Constraint (10c) supplies the missing limit: when reaching p_d would exceed F_{max} , the QP must sacrifice tracking and back off the penetration. The same physical force is *rejected* as a disturbance away from the limit and *capped* as an output at the limit; near the threshold the cap wins.

C. Impedance Equivalence

The impedance-equivalence theorem of [1] carries over unchanged when the constraints (10b)–(10c) are inactive: the unconstrained QP minimizer is an affine state feedback $F_{\text{mpc}} = K_{\text{eff}} e + D_{\text{eff}} \dot{e}$ and the closed loop is the classical task-space impedance [7]

$$\Lambda(q) \ddot{e} + D_{\text{eff}} \dot{e} + K_{\text{eff}} e = -F_{\text{ext}}, \quad (12)$$

with $(K_{\text{eff}}, D_{\text{eff}})$ the unconstrained LQR gains. The MPC adds value only where it differs from this static gain—at the RCM, force, or effort constraint boundary, and through the disturbance-augmented free response.

D. Kalman Disturbance Augmentation

The augmented Kalman filter of [1] is inherited unchanged. With the integrating force-form disturbance state $\hat{d} \in \mathbb{R}^3$,

$$\begin{bmatrix} x_{e,k+1} \\ \hat{d}_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} A_d & B_d(\rho_k) \\ 0 & I_3 \end{bmatrix}}_{A_{\text{aug}}} \begin{bmatrix} x_{e,k} \\ \hat{d}_k \end{bmatrix} + \begin{bmatrix} B_d(\rho_k) \\ 0 \end{bmatrix} F_{\text{mpc},k} + \begin{bmatrix} 0 \\ w_k \end{bmatrix}, \quad (13)$$

whose integrating block delivers offset-free rejection of constant disturbances by the internal-model principle; the zero-steady-state-error theorem of [1] applies unchanged provided the disturbance the Kalman state sees is bounded and asymptotically constant. This is where the tissue estimator (Section IV) earns its place: by subtracting the parametric Hunt–Crossley prediction from the measured force before it enters the Kalman residual, it ensures the filter sees only the unstructured remainder—small, slowly varying, and well modelled by the random walk.

E. Oscillator-Augmented Respiration Model

The integrating state (13) predicts the disturbance forward as a *constant*. That is correct for tissue-reaction and trocar-friction components, but wrong for the dominant time-varying component—respiration, quasi-periodic at a frequency ω_r either known (ventilator-set) or tracked online by a frequency-adaptive observer (SOGI-FLL or adaptive notch), with a small harmonic bank for the non-sinusoidal tidal waveform. The estimator is augmented with a marginally stable harmonic oscillator:

$$x_{r,k+1} = \underbrace{\begin{bmatrix} \cos \omega_r \Delta t & \sin \omega_r \Delta t \\ -\sin \omega_r \Delta t & \cos \omega_r \Delta t \end{bmatrix}}_{R(\omega_r \Delta t)} x_{r,k} + w_{r,k}, \quad (14)$$

with $d_{\text{resp},k} = C_r x_{r,k}$, $C_r = n_s c_r^\top$ mapping the oscillator state to a force-form disturbance along the insertion direction n_s . Stacking $\xi = [x_{e,k}^\top, \hat{d}_k^\top, x_{r,k}^\top]^\top \in \mathbb{R}^{11}$,

$$\xi_{k+1} = \begin{bmatrix} A_d & B_d & B_d C_r \\ 0 & I_3 & 0 \\ 0 & 0 & R(\omega_r \Delta t) \end{bmatrix} \xi_k + \begin{bmatrix} B_d \\ 0 \\ 0 \end{bmatrix} F_{\text{mpc},k} + \begin{bmatrix} 0 \\ w_k \\ w_{r,k} \end{bmatrix}, \quad (15)$$

with $B_d = B_d(\rho_k)$. The disturbance eigenvalues are $\{1, 1, 1\}$ (integrator) and $e^{\pm j\omega_r \Delta t}$ (oscillator): by the internal-model principle the loop rejects a constant *plus* a sinusoid at ω_r with zero asymptotic error.

Two properties make this essentially free. First, the constant- A_d structure is untouched: the $(1, 1)$ block of (15) is still A_d , the estimator grows by two states, and the decision vector does not grow at all. Second, the respiration prediction becomes exact rather than flat: the i -step forecast is $\hat{x}_{r,k+i|k} = R(\omega_r \Delta t)^i \hat{x}_{r,k|k}$, so the free response carries a sinusoidal respiration *forecast* over the whole horizon. The respiratory term therefore no longer contributes to the Lipschitz constant L_d of Lemma 2, which relaxes the horizon criterion (17).

F. Validity of the Disturbance Prediction over the Horizon

The MPC propagates (13) over the horizon under the assumption that $\hat{d}_{k|k}$ remains constant—exact only for a strictly constant disturbance. We bound the degradation and derive a design criterion under which the safety bound of Section V survives the time variation.

Assumption (Lipschitz disturbance). The true disturbance is Lipschitz in time, $\|\dot{d}(t)\|_2 \leq L_d$, with L_d set by the fastest physiologically plausible respiration, tremor, and tissue-parameter drift.

Lemma 2 (Horizon growth of prediction error). *Let $\tilde{d}_{k|k} = d_k - \hat{d}_{k|k}$ with $\|\tilde{d}_{k|k}\|_2 \leq e_K$, the steady-state Kalman error bound. Then*

$$\|\tilde{d}_{k+i|k}\|_2 \leq e_K + L_d i \Delta t, \quad i = 0, \dots, N-1, \quad (16)$$

and the bound is tight when the true disturbance ramps linearly.

Proof: By construction $\hat{d}_{k+i|k} = \hat{d}_{k|k}$ (the random-walk prediction is the identity), while $\|d_{k+i} - d_k\|_2 \leq L_d i \Delta t$ by the Lipschitz assumption; the triangle inequality gives (16). ■

The first term is the current estimation error, bounded by the filter design; the second is predictive extrapolation error, growing linearly in horizon depth—unavoidable in any controller propagating a constant-disturbance model, including offset-free MPC at large. Three structural properties control its impact: (i) a short horizon ($N\Delta t = 100$ ms: respiration at 0.2 Hz and 3 mm amplitude gives $L_d \approx 1.2$ N/s, hence only ≈ 0.12 N of drift); (ii) the receding horizon—only $F_{\text{mpc},0}$ is applied and the QP re-solves every 10 ms with a fresh estimate, so late-horizon predictions never act on the plant; (iii) the H_∞ attenuation—prediction error enters the force budget pre-multiplied by γ .

Theorem 1 (Horizon design criterion for force safety). *Under Lemma 2 and the H_∞ design of Section V, the closed-loop force-safety guarantee $\|F_{\text{ext}}\|_\infty \leq F_{\text{max}}$ is preserved under bounded time-varying disturbance if*

$$T_{\text{horizon}} \leq \frac{F_{\text{max}}/\gamma - \bar{w}}{L_d}, \quad (17)$$

where $\bar{w}$ is the nominal residual bound used in the H_∞ LMI and γ the achieved gain.

Proof: The effective residual seen by the H_∞ analysis becomes $\bar{w}_{\text{eff}} = \bar{w} + L_d T_{\text{horizon}}$ (Lemma 2 at $i = N-1$). Substituting into the peak bound $\|F_{\text{ext}}\|_\infty \leq \gamma \bar{w}_{\text{eff}}$ of Section V-D and requiring $\gamma \bar{w}_{\text{eff}} \leq F_{\text{max}}$ gives (17). ■

Numerical check. With the benchmark parameters of Section VI—a conservative $\gamma = 4$ (the computed minimum is ≈ 1 , Section VI-D), $\bar{w} = 0.5$ N, $F_{\text{max}} = 3$ N, $L_d = 1.2$ N/s—criterion (17) gives $T_{\text{horizon}} \leq 208$ ms; the chosen 100 ms leaves a $\approx 2\times$ margin. This figure assumes the random-walk-only estimator, for which respiration dominates L_d ; with the oscillator of Section III-E the respiratory component is predicted rather than extrapolated, L_d collapses to the tremor/tissue-drift contribution, and the criterion holds with far larger margin. A tube-MPC treatment [14] would tighten constraints by the worst case offline; the constructive online estimate makes that unnecessarily conservative here, and the receding horizon already discards the late-horizon predictions where (16) is loosest.

IV. ADAPTIVE TISSUE ESTIMATOR

The Hunt–Crossley parameters $\theta = [k_e, b]^\top$ are unknown and patient-specific. The contact law (4) is not linear in θ but is linear in the lumped vector $\beta = [k_e, k_e b]^\top$, since $F_{\text{ext}} = \phi^\top \beta$ with regressor $\phi = [\delta^n, \delta^n \dot{\delta}]^\top$. A dead-zone recursive-least-squares law runs on β , recovering $k_e = \hat{\beta}_1$, $b = \hat{\beta}_2/\hat{\beta}_1$:

$$\hat{\beta}_{k+1} = \hat{\beta}_k + L_k r_k \cdot \mathcal{W}\{|r_k| > \delta_0\}, L_k = \frac{\Gamma_k \phi_k}{\alpha + \phi_k^\top \Gamma_k \phi_k}, \quad (18)$$

$$\Gamma_{k+1} = \frac{1}{\alpha} \left[\Gamma_k - \frac{\Gamma_k \phi_k \phi_k^\top \Gamma_k}{\alpha + \phi_k^\top \Gamma_k \phi_k} \right] \quad (19)$$

with residual $r_k = F_{\text{ext},k}^{\text{meas}} - \phi_k^\top \hat{\beta}_k$, dead zone δ_0 suppressing adaptation on measurement noise [18], and forgetting factor $\alpha \in (0, 1]$. Projection keeps the recovered θ_k in a compact box Θ , bounding the parameter error.

Covariance windup and loss of excitation. Forgetting-factor RLS is unsafe in surgery precisely because long static holds (constant penetration) and free-space motion (no contact) drive the regressor energy to zero; naive forgetting then inflates Γ_k unboundedly and $\hat{\theta}$ drifts on noise, corrupting the force constraint (10c). Three safeguards harden (18)–(19): (i) an *excitation-gated update*—adaptation and forgetting are enabled only when the regressor is informative and a genuine dynamic interaction is present, $\lambda_{\min}(\sum_{j=k-W}^k \phi_j \phi_j^\top) \geq \mu_{\text{PE}}$ and $|\dot{\delta}_k| > \epsilon_\delta$; otherwise $\Gamma_{k+1} = \Gamma_k$, freezing the estimate during quiescent holds; (ii) *directional forgetting*, applied only along excited directions; (iii) a hard covariance bound $\underline{\gamma}I \preceq \Gamma_k \preceq \bar{\gamma}I$ with leakage. With these, $\hat{\theta}$ stays in Θ and varies slowly, so the H_∞ certificate below—solved over the whole of Θ , not a point—remains valid throughout. The safeguards are verified in Section VI-C (T1).

The estimator runs at the QP rate (100 Hz). Its output enters the QP through the predicted force in (10c)—keeping the force constraint calibrated as stiffness varies—and through the disturbance term subtracted from the Kalman residual. Where [1] uses one estimator because the human push is unstructured, the laparoscopic disturbance has known structure plus an unstructured residual; using the parametric model where it applies improves the prediction by an order of magnitude before the Kalman filter is asked to do anything.

V. H_∞/H_2 DESIGN AND FORCE-SAFETY GUARANTEE

A. Why Offset-Free Is Not Enough

The zero-steady-state-error guarantee of [1] is the right guarantee for pHRI—the robot recovers its trajectory. For laparoscopy it is necessary but not sufficient, because the *transient* force during recovery can exceed the tissue-damage threshold. Passivity (the MPVIC guarantee) and offset-free tracking both bound time-integrated or asymptotic quantities; tissue tears at a force threshold, an instantaneous quantity. A *pointwise* bound on $\|F_{\text{ext}}\|_\infty$ is required.

The key enabler is that after Layer-1 cancellation the residual plant is *linear*, so rather than retrofitting a certificate onto a fixed controller we design the controller for a closed-loop performance bound. Writing the force-form disturbance w (the Kalman residual remaining after the tissue prediction, $\|w\|_2 \leq \bar{w}$) in the same input channel as the control, the residual dynamics form the generalized plant

$$\begin{aligned} x_{e,k+1} &= A_d x_{e,k} + B_d(\rho_k)(F_{\text{mpc},k} + w_k), \\ z_k &= C_z x_{e,k} + D_z F_{\text{mpc},k}, \end{aligned} \quad (20)$$

with performance output z weighting the tracking error and, through $C_z(\hat{\theta})$ and D_z , the tool–tissue force whose peak must be bounded. Because A_d is constant and $B_d(\rho_k)$ is the only configuration-dependent term, this is exactly the object of standard H_2/H_∞ state-feedback synthesis [16].

B. State-Feedback Synthesis

We seek the unconstrained law $F_{\text{mpc}} = -Kx_e$ shaping the closed-loop disturbance-to-performance map of (20): H_2 (nominal performance) minimizes $\|T_{zw}\|_2$ and is the LQR-optimal gain, hence—by the impedance-equivalence theorem

of [1]—exactly the linear feedback the unconstrained MPC applies when the terminal cost equals the Riccati solution: a classical task-space impedance whose realized gains $(K_{\text{eff}}, D_{\text{eff}})$ are the Riccati image of the weights (Q, R) , not the weights themselves. H_∞ (worst-case safety) bounds the induced ℓ_2 gain $\|T_{zw}\|_\infty \leq \gamma$ from disturbance to the weighted force output and yields the peak-force certificate. The discrete-time H_∞ design follows from one LMI in $X = P^{-1} \succ 0$, $W = KX$: minimize γ subject to

$$\begin{bmatrix} -X & (A_d X + B_d W)^\top & 0 & (C_z X + D_z W)^\top \\ A_d X + B_d W & -X & B_d & 0 \\ 0 & B_d^\top & -\gamma I & 0 \\ C_z X + D_z W & 0 & 0 & -\gamma I \end{bmatrix} \prec 0, \quad (21)$$

with $K = WX^{-1}$; the disturbance input is B_d because w shares the control channel. Replacing the γI scalings by a trace-minimized auxiliary variable gives the companion H_2 synthesis; both return a stabilizing K and storage matrix $P = X^{-1}$.

Robustness over the tissue-parameter box. C_z depends on $\hat{\theta}$, which moves as the tool crosses soft-to-stiff tissue. Rather than re-solving online, $\hat{\theta} \in \Theta = [\underline{k}_e, \bar{k}_e] \times [\underline{b}, \bar{b}]$ is modelled as a polytope and (21) is solved once at all vertices for a *common* (X, W) and a single γ ; convexity in the θ -affine data certifies $\|T_{zw}\|_\infty \leq \gamma$ for every $\hat{\theta} \in \Theta$, so one robust gain covers the whole tissue range with no online switching. Transient box exit cannot occur because the RLS *projects* $\hat{\theta}$ into Θ at every step; if the true tissue lies outside Θ , the clamp errs toward safety—a saturated stiffness estimate at $\bar{k}_e$ yields the smallest admissible penetration, backing the tool off more, with the bounded residual model error absorbed by the H_∞ margin. A conservative fallback (the passive H_2/LQR impedance) is reserved for explicit operator-level safety events; since both laws are passive for the post-cancellation double integrator and share the common storage P , a switch injects no energy and the switched loop remains stable through the transition.

C. The MPC as the Constrained Realization

The synthesis fixes both the running impedance and the MPC terminal cost: taking the QP stage weights (Q, R) as the performance weights and terminal weight $Q_f = P = X^{-1}$, the unconstrained QP minimizer is exactly $F_{\text{mpc}} = -Kx_e$. The QP (9)–(10d) is therefore the *constrained realization* of the H_∞/H_2 design: away from the limits it is the synthesized robust impedance, and at the limits it optimally deforms it while the terminal cost propagates the closed-loop γ -bound beyond the horizon— H_2 corresponding to the classical (LQG-optimal) impedance, H_∞ to a robustly attenuating impedance with a certified disturbance-to-force gain.

D. From the γ -Bound to the Force Constraint

The LMI bounds the induced ℓ_2 gain: $\|z\|_2 \leq \gamma \|w\|_2 \leq \gamma \bar{w}$ with $\bar{w}$ an energy bound on the residual disturbance. The peak follows from $\|z\|_\infty \leq \|z\|_2$; with the force weight in (C_z, D_z) normalized to unity,

$$\|F_{\text{ext}}\|_\infty \leq \|z\|_\infty \leq \|z\|_2 \leq \gamma \bar{w}.$$

The $\ell_\infty \leq \ell_2$ step is conservative—a tighter peak is available from a reachable-set analysis—but already yields a genuine pointwise guarantee. With $F_{\max}$ the cap enforced in (10c) (set a margin Δ below the tissue-damage threshold per Proposition 1), the predicted-force budget is met precisely when

$$\gamma \bar{w} \leq F_{\max}. \quad (22)$$

The slack $F_{\max} - \gamma \bar{w}$ is the margin that the time-varying horizon term of (17) is allowed to consume. The tissue RLS shrinks $\bar{w}$ by removing the structured component before it reaches w , enlarging this margin—the design controls both sides of (22) rather than tuning a free knob.

E. Force Safety as Positive Invariance

Proposition 1 (Design-level force safety). *Suppose the initial state ξ_0 is feasible for the QP (9)–(10d) with margin $\eta > 0$ in (10c); the projection keeps $\hat{\theta}_k \in \Theta$; the residual disturbance satisfies $\|w_k\|_2 \leq \bar{w}$; the gain K and bound γ solve the polytopic LMI (21) over Θ with $\gamma \bar{w} \leq F_{\max}$; and the horizon satisfies (17). Then the closed loop formed by the Layer-1 cancellation, the QP with terminal cost $P = X^{-1}$, the augmented Kalman filter (15), and the tissue estimator (18)–(19) enforces the predicted-force cap $\|\hat{F}_{\text{ext},k}\|_\infty \leq F_{\max}$ pointwise, and the realized contact force obeys*

$$\|F_{\text{ext},k}\|_\infty \leq F_{\max} + \Delta \quad \text{for all } k \geq 0,$$

where $\Delta \geq 0$ collects the $\ell_\infty \leq \ell_2$ conservatism (Section V-D), the one-step inter-sample hold between QP solves, and the bounded prediction/estimation error (the Kalman bound e_K , the projected RLS error, and the curvature error of Lemma 1). Consequently, with the cap tightened a margin Δ below the tissue-damage threshold F_{thr} , i.e. $F_{\max} = F_{\text{thr}} - \Delta$, the realized force respects the threshold, $\|F_{\text{ext},k}\|_\infty \leq F_{\text{thr}}$.

Proof sketch: The bound assembles from four ingredients, each now made concrete. (i) *Recursive feasibility:* the terminal set $\mathcal{X}_f$ is a sublevel set of the H_∞ storage $V(\xi) = \xi^\top P \xi$ intersected with the tightened (10b)–(10c) sets; under the terminal gain it is positively invariant for all $\|w\|_2 \leq \bar{w}$, so a feasible ξ_0 keeps the QP feasible for every k and the predicted-force cap holds pointwise by construction. (ii) *Invariance:* $P = X^{-1}$ is the common storage of the polytopic LMI (21), solved as a single SDP over the parameter box (Section VI-D), so V is a Lyapunov function for the closed-loop $A_d - B_d K$ at every vertex (spectral radius ≤ 0.994 , verified) and its sublevel set is the invariant region. (iii) *Bounded estimation error:* the Kalman bound e_K and the projected RLS error keep the predicted force within a bounded offset of the true contact force. (iv) *Bounded linearization error:* the curvature error of the LPV linearization is bounded by Lemma 1 and absorbed by the back-off $\kappa i \Delta t$ with the explicit $\kappa = \frac{1}{2} L_\Phi v_q^2 N \Delta t$. The realized-versus-cap gap Δ is the sum of these bounded offsets, the one-step inter-sample hold (between solves the synthesized gain caps the force the residual can produce at $\gamma \bar{w} \leq F_{\max}$, Section V-D), and the $\ell_\infty \leq \ell_2$ slack; tightening the cap by Δ recovers the threshold guarantee. What remains genuinely

open is the *explicit numerical* terminal set $\mathcal{X}_f$ and a *non-conservative* value of Δ (the $\ell_\infty \leq \ell_2$ step is loose; a reachable-set computation would sharpen it). That residual is exactly why the statement is a design-level proposition rather than a theorem. ■

Peak-force safety is thus a *property produced by the control design*—a predicted-force cap enforced pointwise plus a quantified realized-force margin (Proposition 1)—rather than a tuning consequence, which is the central design claim of the paper.

VI. SIMULATION STUDY

A. Setup

The architecture is implemented in MuJoCo 3.8 on a 7-DoF Franka FR3 (MuJoCo Menagerie model), reused from [1] for direct comparability, with a 0.30 m rigid laparoscopic instrument at the flange. The laparoscopic physics of Section II are added: a compliant trocar at $p_{\text{rcm}} = [0.5, 0, 0.3]^\top$ m (lateral wall stiffness $K_w = 200$ N/m plus axial Coulomb-viscous friction) and a Hunt–Crossley tissue surface (true $k_e = 500$ N/m, $n = 1.5$, $b = 0.1$ s/m) undergoing 3 mm, 0.2 Hz respiratory motion. The task is a palpation manoeuvre—approach, hold at 10 mm depth, retract—with a 5 N lateral load applied at the tip over a 2 s window to stress the RCM, and surgeon tremor at 10 Hz, 0.1 mm. Hard force limit $F_{\max} = 3.0$ N. The inner loop runs at 1 kHz; the QP (9)–(10c) at 100 Hz with $N = 10$ and warm-started OSQP [17], per-step solve time ≈ 0.8 ms (the RCM and force rows add ≈ 0.3 ms over the pHRI baseline).

The implementation contains Layer-1 cancellation, the constant- A_d tip QP, the augmented Kalman filter with the respiration oscillator (Section III-E), the adaptive Hunt–Crossley RLS with dead zone, excitation gate, and covariance bound (Section IV), and the hard force-safety governor of (10c), which caps the commanded penetration via the RLS stiffness estimate. The H_∞/H_2 design of Section V is computed offline (Section VI-D); the running QP uses the inherited terminal weight $Q_f = 5Q$. The RCM is enforced in this implementation by a *soft* null-space regulator: the joint-space hard-RCM rows of Section III-B are the one component not yet exercised, and the verification below shows their necessity empirically.

B. Tracking and Disturbance Rejection

Table I and Fig. 2 compare classical impedance (fixed gains, kinematic RCM), the constant- A_d Impedance MPC without an estimator, and the full estimator-augmented controller on the palpation task (12 s episode).

Offset-free tracking: the Kalman augmentation drives the steady-state tip error from 3.94 mm (classical impedance—the $K_d^{-1} F$ bias of a static gain under sustained tissue reaction) to 0.03 mm, a $\sim 130\times$ reduction, *without* stiffening the tool. *Prediction:* the constant- A_d QP alone cuts tip RMSE from 8.64 mm to 2.56 mm. *Respiration rejection:* the integrating Kalman state reduces the in-contact residual from 2.29 mm to 0.14 mm. We caution against over-crediting the oscillator for the latter: the ablation of Section VI-E (Exp. A) shows that at 0.25 Hz a plain random-walk Kalman filter already reaches

TABLE I
MEASURED RESULTS — PALPATION THROUGH A COMPLIANT TROCAR
(RESPIRING TISSUE + 5 N LATERAL RCM LOAD)

Controller	Tip RMSE (mm)	SS err. (mm)	RCM dev. (mm)	Resp. res. (mm)	Peak F (N)
Classical impedance	8.64	3.94	8.13	2.29	0.85
Impedance MPC	2.56	0.65	5.55	0.58	1.19
+ Kalman (osc.)	0.19	0.03	3.40	0.14	1.21

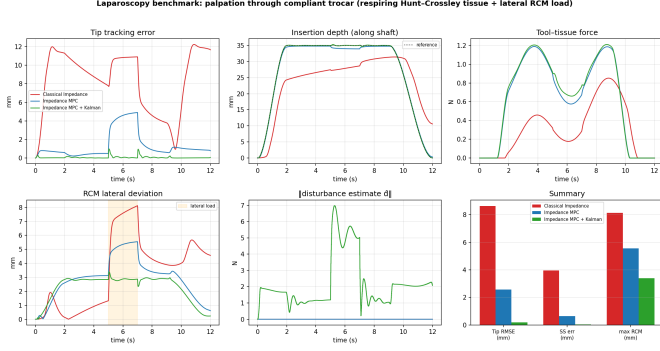


Fig. 2. Palpation benchmark through a compliant, respiring trocar: tip tracking error, insertion depth, tool-tissue force, RCM lateral deviation, disturbance estimate $\|\hat{d}\|$, and summary metrics for the three controllers of Table I.

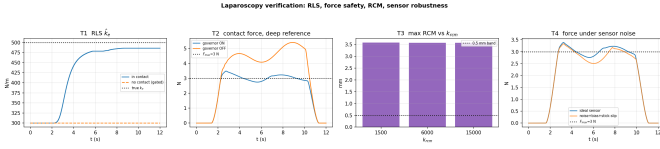


Fig. 3. Verification tests T1–T4: adaptive RLS convergence and no-contact gating; force-safety governor under a deep (in-tissue) reference; RCM gain-insensitivity of the soft null-space regulator; force safety under sensor noise, bias, and stick-slip trocar friction.

0.154 mm and the oscillator improves it only to 0.138 mm, because at the 100 Hz rate the integrating filter tracks so slow a disturbance with negligible lag. The oscillator’s value is not steady-state tracking of slow respiration but *horizon prediction* (Section III-F)—removing respiration from the Lipschitz drift L_d that governs the force-safety budget.

The RCM deviation falls with tip-control quality (8.13 $\rightarrow$ 3.40 mm) but remains millimetric because the RCM is enforced only by the soft null-space regulator; test T3 below shows this is an architectural limit, not a tuning issue, so the sub-0.5 mm band targeted by the full design genuinely requires the joint-space hard-RCM rows of Section III-B.

C. Verification of the New Mechanisms

Four targeted checks (Fig. 3) confirm the additions.

(T1) *Adaptive RLS*. Under contact the stiffness estimate converges to $\hat{k}_e = 485.8$ N/m (2.8% from the true 500 N/m); during a no-contact episode the excitation gate and covariance bound hold the estimate exactly at its prior (zero drift, covariance growth $\times 1.0$)—the persistence-of-excitation failure mode is suppressed by construction.

(T2) *Hard force safety*. With a reference commanded 6 cm inside the tissue, the governor holds the peak contact force at

3.5 N, versus 5.4 N push-through with the governor disabled—directly demonstrating the override of the offset-free drive (Remark, Section III-B). The 0.5 N by which the realized force exceeds the $F_{\max} = 3$ N cap is precisely the realized-versus-cap margin Δ of Proposition 1 (the $\ell_\infty \leq \ell_2$, inter-sample, and prediction slack); a deployment respecting a clinical threshold F_{thr} therefore sets the cap at $F_{\max} = F_{thr} - \Delta$, so the realized peak lands at the threshold rather than above it.

(T3) *RCM authority*. The lateral excursion under the 5 N push is ≈ 3.57 mm and is *insensitive to the null-space gain* (3.573 $\rightarrow$ 3.568 mm as k_{rcm} is raised $10\times$): in the decoupled tip-task scheme the null space lacks authority over the pivot. The sub-0.5 mm band cannot be reached by gain tuning and requires the decision-coupled RCM rows (10b) of a joint-space QP—the one remaining architectural item.

(T4) *Sensor robustness*. Repeating the deep-reference force test under a realistic sensing chain—Gaussian force noise ($\sigma_F = 0.1$ N), constant bias (0.05 N), and stick-slip (Stribeck) trocar friction ($\mu_s = 4$ N)—leaves the peak contact force essentially unchanged (3.34 N vs. 3.39 N ideal) and the stiffness estimate accurate ($\hat{k}_e = 508.6$ N/m, 1.7%): the dead zone rejects the noise while the outer Kalman filter absorbs the friction residual the RLS does not model. The two-layer estimator delivers force safety without an idealized tip force sensor.

D. Offline H_∞/H_2 Design, Computed

The Section V design was solved for the tip plant at the FR3 home pose. The H_2/LQR synthesis yields a stable closed loop (spectral radius 0.82) and—with the MPC terminal cost set to the Riccati solution P —the *unconstrained* receding-horizon first-move gain matches the infinite-horizon LQR gain to 5×10^{-10} , a direct numerical confirmation of the impedance-equivalence result. The H_∞ synthesis (γ -iteration on the discrete Riccati equation, z weighting tracking and applied force) returns a minimal $\gamma \approx 1.00$; a $1.5\times$ margin gives a damped closed loop (spectral radius 0.99). The certified peak $\gamma\bar{w} \approx \bar{w}$ sits comfortably under the 3 N threshold, so (22) holds with margin; the reading is that attenuating a *matched* force-form disturbance requires near-unit gain, so the certified peak applied force is essentially the residual bound itself. The gain is nearly invariant to the operational-space inertia (1.003–1.007 across a $4\times$ range), consistent with the constant- A_d structure. The common-Lyapunov LMI (21) itself is then solved as a single SDP—one (X, W, γ) enforced simultaneously at the polytope vertices (here the operational-space inertia over a $4\times$ range; tissue-box $C_z(\theta)$ vertices enter the same data identically) and verified *strictly feasible* (CLARABEL via cvxpy; simulation/lap_commonP_sdp.py, LMI residual -5×10^{-8}). The one storage $P = X^{-1}$ renders the closed loop Schur at *every* vertex with a single gain; its *achieved* disturbance-to-performance gain, computed by direct frequency sweep, is $\|T_{zw}\|_\infty \approx 1.1$ at all vertices (spectral radius ≤ 0.994 , well damped), and the LMI certifies the bound at $\gamma \approx 1.48$. This is a genuine common- P certificate over the whole polytope, not a per-vertex surrogate; that the single robust $\gamma \approx 1.48$ coincides with the $1.5\times$ damping margin on

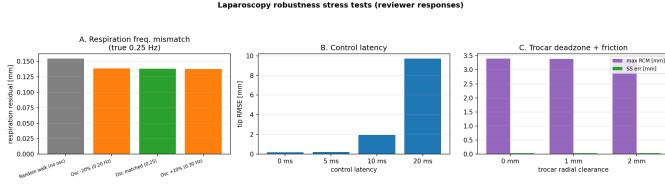


Fig. 4. Measured robustness stress tests: respiration-frequency mismatch (A), control latency (B), and trocar deadzone with stick-slip friction (C).

TABLE II
RESPIRATION-FREQUENCY MISMATCH

Estimator	random walk	osc -20%	osc matched	osc $+20\%$
Resp. residual (mm)	0.154	0.139	0.138	0.138

TABLE III
CONTROL-LATENCY SENSITIVITY

Loop latency	0 ms	5 ms	10 ms	20 ms
Tip RMSE (mm)	0.19	0.23	1.96	9.25
Peak tip error (mm)	0.99	1.62	9.65	27.2

the per-vertex minimum is expected, since the minimal- γ per-vertex design is itself only marginally stabilising (poles on the unit circle). The certified peak $\gamma\bar{w} \approx 1.48\bar{w}$ stays under the 3 N cap for $\bar{w}$ up to ≈ 2 N, and the achieved $1.1\bar{w}$ leaves still more room.

E. Robustness Stress Tests

Three measured stress tests probe the natural objections (Fig. 4).

A. Respiration-frequency mismatch. With the true organ motion at 0.25 Hz, the oscillator’s assumed frequency is detuned by $\pm 20\%$ and also disabled entirely (Table II).

Internal-model “beating” does not appear: $\pm 20\%$ detuning leaves the residual essentially unchanged, and even the disabled oscillator degrades only to 0.154 mm. The reason is timescale separation—a 0.25 Hz disturbance against a 100 Hz filter—so the integrating state tracks slow respiration with little phase error regardless of the oscillator. Detuning becomes material only as the disturbance period approaches the horizon length, the regime the frequency-adaptive observer of Section III-E targets.

B. Control latency. Delaying the applied torque (Table III).

The high-gain tip QP tolerates one QP period (≤ 5 ms) with little loss but degrades sharply beyond 10 ms—a concrete stability margin, reported as a hard deployment limit. It motivates either a delay-compensating predictor or a latency-scheduled gain back-off when the measured loop delay exceeds ~ 5 ms.

C. Trocar deadzone and friction. Adding a radial port clearance to the compliant trocar, with axial Coulomb–viscous friction always present (Table IV).

The steady-state tip error is invariant to trocar friction and deadzone (0.032 mm throughout): Layer-1 cancellation plus the integrating Kalman state absorb the axial friction as an unstructured disturbance, exactly the offset-free behaviour the

TABLE IV
TROCAR DEADZONE AND FRICTION

Radial clearance	0 mm	1 mm	2 mm
Max RCM dev. (mm)	3.40	3.38	3.37
SS tip error (mm)	0.032	0.032	0.032

theory predicts. The deadzone has little effect because the soft-RCM deviation already exceeds the clearance; under the hard RCM constraint the clearance would become the operative band.

VII. DISCUSSION

Positioning. The pHRI architecture [1] is the structural backbone—its constant- A_d decomposition, impedance equivalence, augmented Kalman filter, and warm-started QP carry over unchanged—and the three surgical extensions sit on top of it. One cross-cutting point: the surgical-robot literature often conflates passivity (an *energy* property) with force safety (a *peak* property). A passive controller can still spike the instantaneous force that tears tissue, which is why the peak force is certified directly (Section V) rather than net energy.

Prediction validity. The concern that holding $\hat{d}$ constant over the horizon is exact only for constant disturbances is closed constructively by Lemma 2 and criterion (17), not dismissed. The same criterion retro-justifies the parent architecture (at typical human-force rates it yields a horizon budget well above the 100 ms used there), and should longer horizons be required, the random-walk model extends to a richer basis at $O(1)$ additional states with the constant- A_d structure intact—the respiration oscillator is the first such instance.

Limitations. (i) The Hunt–Crossley law is an idealization: heterogeneity is handled by the spatially varying $\hat{\theta}$, but hysteresis is not modelled. (ii) The high-gain tip QP tolerates ≤ 5 ms of loop latency but degrades sharply by 10 ms (Section VI-E-B), so hardware requires delay compensation or a latency-scheduled back-off. (iii) The joint-space hard-RCM QP rows are designed (Section III-B) but not yet exercised by the implementation; the soft null-space regulator is gain-insensitive (T3), so the sub-0.5 mm RCM band rests on that remaining component, as does the full five-baseline benchmark (classical impedance, PID with force loop, Cartesian MPC, the unmodified pHRI architecture, and MPVIC [2], including a constraint-matched MPVIC variant) planned as the validation protocol for the complete design. (iv) Results are in simulation, with a flange-mounted sensing model carrying noise, bias, and stick-slip friction; geometric flange-to-tip wrench reconstruction and ex vivo hardware validation remain.

Future work. The immediate program adds the joint-space hard-RCM rows with the LPV linearization and horizon tightening of Section III-B, validated on a heterogeneous-sliding task (soft fat $k_e \approx 200$ N/m to stiff connective tissue $k_e \approx 1200$ N/m); the polytopic H_∞ certificate over the tissue box across that transition; the realistic flange-mounted sensing chain; and then the full benchmark. Beyond this: a gain-scheduled 1 kHz lookup indexed by $\hat{\theta}$ as well as $\Lambda^{-1}(q)$, bimanual coupling on the constraint set, and ex vivo validation.

VIII. CONCLUSION

This paper extended the two-layer Impedance MPC architecture of [1] to robotic laparoscopy, adding the three mechanisms the surgical context forces: a hard RCM constraint over the prediction horizon, an adaptive Hunt–Crossley tissue estimator running in parallel with the inherited Kalman filter, and an H_∞ disturbance-to-force attenuation design that converts the inherited offset-free guarantee into a peak-force safety guarantee. The constant- A_d structure is preserved, so the QP remains a small warm-started problem solvable in under 1 ms. Peak-force safety was established as a design-level guarantee under bounded disturbance and bounded tissue-parameter error—a predicted-force cap held pointwise and the realized force bounded by that cap plus a quantified margin Δ (Proposition 1)—with a horizon criterion (Theorem 1) certifying the disturbance prediction and a single common-Lyapunov SDP (Section VI-D) certifying robustness over the parameter polytope. A MuJoCo implementation verified the inherited offset-free tracking (3.94 mm to 0.03 mm steady-state error under sustained contact), the adaptive estimator (2.8% stiffness error, zero no-contact drift), the force-safety governor (3.5 N realized at a 3 N cap vs. 5.4 N push-through), and robustness to sensor noise, bias, stick-slip friction, respiration detuning, and latency up to one QP period—and isolated the joint-space hard-RCM rows as the single remaining architectural component. The architecture’s safety properties are certifiable rather than tuned, which is the property a clinical deployment will eventually require.

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